

Sequences and Series 1

By Joshua Aaron

April 2018

Small Steps: Here we will go through the steps necessary to calculate, and hopefully understand our first infinite series. To start though, we will go through some notation:

1. Suppose the following is true:

$$\sum_{k=1}^3 \frac{1}{2^k} = \frac{1}{2^1} + \frac{1}{2^2} + \frac{1}{2^3} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

How would you expand the following as a sum of multiple terms, what is the result?

$$\sum_{k=4}^5 \frac{1}{2^k}$$

2. How would I write the following using “Sigma Notation?” (This symbol, Σ is the (Uppercase) Greek letter, *Sigma*, we use it to refer to a Ssum.)

$$\frac{1}{3^1} + \frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4}$$

3. We can write the sum of the integers from 100 to 200 as

$$\sum_{k=100}^{200} k$$

How might we write the sum of all integers between a and b ? What conditions might we want to put on a and b ? What would the sum of all integers from a to $-a = -1 \cdot a$ be?

4. Let’s flip it around, based on the above, how might we write 10 as a sum of consecutive integers?
5. How about 15?
6. How about 9?
7. Can you write 0 as the sum of 5 consecutive integers? (might want to start with a negative!)

Small Acts of Translation: Try writing out the following in Sigma notation and as an extended sum:

1. The sum of the first 100 numbers
2. The sum of the first four terms of a geometric series with common ratio 4, starting with 2
3. The sum of the first 8 powers of 2
4. The sum of the first 8 powers of $\frac{1}{2}$
5. The sum of the first four powers of $\frac{2}{3}$ minus the first four powers of $\frac{1}{6}$